

1. Arithmetic

QUESTIONS 31 / 31 completed

Arithmetic

ARITHMETIC

Write a Java program that takes two integers as input from the user and calculates their sum, difference, product, and quotient using arithmetic operators.

PROGRAM EDITOR

Java Run Save

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         int a=sc.nextInt();
6         int b=sc.nextInt();
7         System.out.println("Sum = "+(a+b));
8         System.out.println("Difference = "+(a-b));
9         System.out.println("Product = "+(a*b));
10        System.out.println("Quotient = "+(a/b));
11    }
12 }
```

OUTPUT

```
Sum = 30
Difference = 10
Product = 200
Quotient = 2
```

INPUT

```
20
10
```

2. Max

QUESTIONS 31 / 31 completed

MAX

MAX

Write a Java program to find the maximum of three numbers using conditional statements.

PROGRAM EDITOR

Java Run Save

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner s=new Scanner(System.in);
5         int a=s.nextInt();
6         int b=s.nextInt();
7         int c=s.nextInt();
8         if(a>b && a>c){
9             System.out.println("a is greater than b & c."+"The value of a is "+a);
10        }
11        else if(b>a && b>c){
12            System.out.println("b is greater than a and c."+"The value of b is "+b);
13        }
14        else{
15            System.out.println("c is greater than a and b."+"The Value of c is "+c);
16        }
17    }
18 }
```

OUTPUT

```
a is greater than a and b.The Value of a is 30
```

INPUT

```
10
20
30
```

3. Leap Year

QUESTIONS 31 / 31 completed

Leap

LEAP
Write a Java program to check whether a given year is a leap year or not using conditional statements.

PROGRAM EDITOR

Java Run Save

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner z=new Scanner(System.in);
5         int a=z.nextInt();
6         if(a%4==0 && a%100!=0){
7             System.out.println(a+" is leap year");
8         }
9         else if(a%400==0){
10            System.out.println(a+" is leap year");
11        }
12        else{
13            System.out.println(a+" is Not a leap Year");
14        }
15    }
16 }
```

OUTPUT

2024 is leap year

INPUT

2024

4. Fibonacci Series

QUESTIONS 31 / 31 completed

Fibonacci

FIBONACCI
Write a Java program to print the Fibonacci series up to a given number using loops.

PROGRAM EDITOR

Java Run Save

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner z=new Scanner(System.in);
5         int a=0;
6         int b=1;
7         int c;
8         int q=z.nextInt();
9         for(int i=1;i<=q;i++){
10            System.out.print(a+" ");
11            c=a+b;
12            a=b;
13            b=c;
14        }
15    }
16 }
```

OUTPUT

0 1 1 2 3

INPUT

5

5. Factorial

QUESTIONS 31 / 31 completed

Factorial

FACTORIAL

Write a Java program to calculate the factorial of a given number using loops.

PROGRAM EDITOR

1000

Run Save

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner s=new Scanner(System.in);
5         int a=s.nextInt();
6         int result=1;
7         for(int i=1;i<=a;i++){
8             result=result*i;
9         }
10        System.out.println("The factorial of a = "+result);
11    }
12 }
```

OUTPUT

The factorial of a = 120

INPUT

5

6.RT-Pattern

QUESTIONS 31 / 31 completed

RT-Pattern

RT-PATTERN

Write a Java program to print the following pattern using loops:
**

PROGRAM EDITOR

1000

Run Save

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner s=new Scanner(System.in);
5         int a=s.nextInt();
6         for(int i=1;i<=a;i++){
7             for(int j=1;j<=i;j++){
8                 System.out.print("**");
9             }
10            System.out.println();
11        }
12    }
13 }
```

OUTPUT

**

INPUT

5

7. Binary Search

QUESTIONS: 31 / 31 completed

Search

SEARCH

Write a Java program to implement binary search on an array of integers using loops and conditional statements.

PROGRAM EDITOR

Java Run Save

```
1  int a[]=new int[n];
2  int[] a=new int[n];
3  boolean found=false;
4  for(int i=0;i<n;i++){
5      a[i]=r.nextInt();
6  }
7  int binary=r.nextInt();
8  for(int i=0;i<n;i++){
9      if(binary==a[i]){
10         found=true;
11         break;
12     }
13 }
14 if(found){
15     System.out.println(binary+" Found");
16 }
17 else{
18     System.out.println(binary+" not found");
19 }
20 }
```

OUTPUT

4 Found

INPUT

4
3 2 4 5
4

8. GCD

QUESTIONS: 31 / 31 completed

GCD

GCD

Write a Java program to find the GCD (Greatest Common Divisor) of two numbers using loops and conditional statements.

PROGRAM EDITOR

Java Run Save

```
1  import java.util.Scanner;
2  public class Main{
3      public static void main(String[] args){
4          Scanner r=new Scanner(System.in);
5          int a=r.nextInt();
6          int b=r.nextInt();
7          int gcd=1;
8          for(int i=1;i<=b;i++){
9              if(a%i==0 && b%i==0){
10                 gcd=i;
11             }
12         }
13         System.out.println(gcd);
14     }
15 }
```

OUTPUT

6

INPUT

12
18

9. Prime

QUESTIONS 25 / 31 completed

Prime

PRIME

Write a Java program to check whether a given number is a prime number or not using loops and conditional statements.

PROGRAM EDITOR

1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16

1 import java.util.Scanner;
2 public class Main{
3 public static void main(String[] args){
4 Scanner s=new Scanner(System.in);
5 int a=s.nextInt();
6 int i;
7 for(i=2;i<=a;i++){
8 if(a%i==0)
9 break;
10 }
11 if(a>1 && i==a)
12 System.out.println("Prime");
13 else
14 System.out.println("Not Prime");
15 }
16 }

OUTPUT

Prime

INPUT

7

10. Sort

QUESTIONS 31 / 31 completed

Sort

SORT

Write a Java program to implement selection sort on an array of integers using loops and conditional statements.

PROGRAM EDITOR

1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24

1 Scanner s=new Scanner(System.in);
2 int n=s.nextInt();
3 int[] a=new int[n];
4 for(int i=0;i<n;i++){
5 a[i]=s.nextInt();
6 }
7 for(int i=0;i<n-1;i++){
8 int min=i;
9 for(int j=i+1;j<n;j++){
10 if(a[j]<a[min])
11 min=j;
12 }
13 int temp=a[i];
14 a[i]=a[min];
15 a[min]=temp;
16 }
17 for(int i=0;i<n;i++){
18 System.out.println(a[i]+" ");
19 }
20 }
21
22
23
24 }

OUTPUT

1
2
3
4
5

INPUT

5
5 4 3 2 1

11. Table

QUESTIONS 25 / 31 completed

Table

TABLE
Write a java program to print the multiplication table of a given number using loops.

PROGRAM EDITOR

1 import java.util.Scanner;
2 public class Main{
3 public static void main(String[] args){
4 Scanner z=new Scanner(System.in);
5 int a=z.nextInt();
6 for(int i=1;i<=a;i++){
7 System.out.println("The Multiplication of "+a+"**"+i+"="+a*i);
8 }
9 }
10 }

Run Save

INPUT

5

OUTPUT

The Multiplication of 5*1=5
The Multiplication of 5*2=10
The Multiplication of 5*3=15
The Multiplication of 5*4=20
The Multiplication of 5*5=25

12. Factorial

QUESTIONS 31 / 33 completed

Factorial

FACTORIAL
2. Write a java program to find the factorial of a given number using recursion.

PROGRAM EDITOR

1 import java.util.Scanner;
2 public class Main{
3 static int fact(int a){
4 int result=1;
5 for(int i=1;i<=a;i++){
6 result=result*i;
7 }
8 return result;
9 }
10 public static void main(String[] args){
11 Scanner z=new Scanner(System.in);
12 int a=z.nextInt();
13 System.out.println(fact(a));
14 }
15 }

Run Save

INPUT

5

OUTPUT

120

13. LT-Pattern

QUESTIONS: 31 / 31 completed

LT-Pattern

LT-PATTERN

Write a Java program to print the following pattern using loops:*****

**
*

PROGRAM EDITOR

Java Run Save

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner s=new Scanner(System.in);
5         int a=s.nextInt();
6         for(int i=5;i>=1;i--){
7             for(int j=1;j<=i;j++){
8                 System.out.print("**");
9             }
10            System.out.println();
11        }
12    }
13 }
```

OUTPUT

```
*****
***
**
*
```

INPUT

5

14. LCM

QUESTIONS: 31 / 31 completed

LCM

LCM

Write a Java program to find the LCM (Least Common Multiple) of two numbers using loops and conditional statements.

PROGRAM EDITOR

Java Run Save

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner s=new Scanner(System.in);
5         int a=s.nextInt();
6         int b=s.nextInt();
7         int lcm=(a*b)/hcf;
8         while(true){
9             if(lcm%a==0 && lcm%b==0){
10                 System.out.println(lcm);
11                 break;
12             }
13             lcm++;
14         }
15     }
16 }
```

OUTPUT

```
36
```

INPUT

12
18

15. D to B

QUESTIONS 31 / 31 completed

D to B

D TO B

Write a Java program to convert a decimal number to binary using loops and conditional statements.

PROGRAM EDITOR

Java Run Save

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner s=new Scanner(System.in);
5         int a=s.nextInt();
6         String binary="";
7         while(a>0){
8             binary=(a%2)+binary;
9             a=a/2;
10        }
11        System.out.println(binary);
12    }
13 }
```

OUTPUT

1101

INPUT

13

16. B to D

QUESTIONS 31 / 31 completed

B to D

B TO D

Write a Java program to convert a binary number to decimal using loops and conditional statements.

PROGRAM EDITOR

Java Run Save

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner s=new Scanner(System.in);
5         int bin=s.nextInt();
6         int dec=0,base=1;
7         while(bin>0){
8             int rem=bin%10;
9             dec=dec+rem*base;
10            base=base*2;
11            bin=bin/10;
12        }
13        System.out.println(dec);
14    }
15 }
```

OUTPUT

13

INPUT

1101

17. Sum of Digits

QUESTIONS: 31 / 31 completed

Sum

SUM

Write a Java program to calculate the sum of the digits of a given number using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner z=new Scanner(System.in);
5         int a=z.nextInt();
6         int sum=0;
7         for(int i=1;i<=a;i++){
8             sum=sum+i;
9         }
10        System.out.println("The Sum of Digit = "+sum);
11    }
12 }
```

OUTPUT

The Sum of Digit = 55

INPUT

10

18. Product of Digit

QUESTIONS: 31 / 31 completed

Product

PRODUCT

Write a Java program to calculate the product of the digits of a given number using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner z=new Scanner(System.in);
5         int a=z.nextInt();
6         int product=1;
7         for(int i=1;i<=a;i++){
8             product=product*i;
9         }
10        System.out.println("The Product of the Digit = "+product);
11    }
12 }
```

OUTPUT

The Product of the Digit = 3628800

INPUT

10

19. Power

QUESTIONS 31 / 35 completed

Power

POWER

Write a Java program to find the power of a number using loops and conditional statements.

PROGRAM EDITOR

Java Run Save

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner s=new Scanner(System.in);
5         int base=s.nextInt();
6         int power=s.nextInt();
7         int result=1;
8         if(power==0){
9             System.out.println(1);
10        }
11        else{
12            for(int i=1;i<=power;i++){
13                result=result*base;
14            }
15            System.out.println(result);
16        }
17    }
18 }
```

OUTPUT

8

INPUT

20. Area of Circle

QUESTIONS 31 / 35 completed

Area Circle

AREA CIRCLE

Write a Java program to calculate the area of a circle using the radius input by the user.

PROGRAM EDITOR

Java Run Save

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner s=new Scanner(System.in);
5         int radius=s.nextInt();
6         double pi=3.14;
7         double Area=pi*(radius*radius);
8         System.out.println("The Area of Circle = "+Area);
9     }
10 }
```

OUTPUT

The Area of Circle = 78.5

INPUT

5

21. Volume of Sphere

QUESTIONS 31 / 31 completed

Shpere

SHPERE

Write a Java program to calculate the volume of a sphere using the radius input by the user.

PROGRAM EDITOR

java Run Save

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner s=new Scanner(System.in);
5         int radius=s.nextInt();
6         double pi=3.14;
7         double q=4.0/3.0;
8         double sphere=q*pi*radius*radius*radius;
9         System.out.println("The volume of Sphere = "+sphere);
10    }
11 }
```

OUTPUT

```
The volume of Sphere = 113.03999999999998
```

INPUT

3

22. Palindrome

QUESTIONS 31 / 31 completed

Palindrome

PALINDROME

Write a Java program to check whether a given string is a palindrome or not using loops and conditional statements.

PROGRAM EDITOR

java Run Save

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner s=new Scanner(System.in);
5         int a=s.nextInt();
6         int temp=a;
7         int rev=0;
8         while(a!=0){
9             int rem=a%10;
10            rev=rev*10+rem;
11            a=a/10;
12        }
13        if(temp==rev){
14            System.out.println("Palindrome");
15        }
16        else{
17            System.out.println("Not a palindrome");
18        }
19    }
20 }
```

OUTPUT

```
not a palindrome
```

INPUT

122

23. Armstrong

QUESTIONS 31 / 31 completed

Armstrong

ARMSTRONG

Write a Java program to check whether a given number is an Armstrong number or not using loops and conditional statements.

PROGRAM EDITOR

Java Run Save

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner s=new Scanner(System.in);
5         int a=s.nextInt();
6         int temp=a;
7         int rev=0;
8         while(a!=0){
9             int rem=a%10;
10            rev=rev+(rem*rem*rem);
11            a=a/10;
12        }
13        if(temp==rev){
14            System.out.println("Armstrong Number");
15        }
16        else{
17            System.out.println("Not a Armstrong Number");
18        }
19    }
20 }
```

OUTPUT

Not a Armstrong Number

INPUT

154

24. RT-num.Pattern

QUESTIONS 31 / 31 completed

RT-num.Pattern

**RT-
NUM.PATTERN**

Write a Java program to print the following pattern using loops:
1
22
333
4444
55555

PROGRAM EDITOR

Java Run Save

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner s=new Scanner(System.in);
5         int a=s.nextInt();
6         for(int i=1;i<=a;i++){
7             for(int j=1;j<=i;j++){
8                 System.out.print(i);
9             }
10            System.out.println();
11        }
12    }
13 }
```

OUTPUT

1
22
333
4444
55555

INPUT

5

25. Perfect

QUESTIONS : 31 / 31 completed

Perfect

PERFECT
Write a Java program to check whether a given number is a perfect number or not using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner s=new Scanner(System.in);
5         int a=s.nextInt();
6         int sum=0;
7         for(int i=1;i<a;i++){
8             if(a%i==0){
9                 sum=sum+i;
10            }
11        }
12        if(a==sum){
13            System.out.println("Perfect Number");
14        }
15        else{
16            System.out.println("Not a Perfect Number");
17        }
18    }
19 }
```

OUTPUT

Not a Perfect Number

INPUT

7

26. Max Array

QUESTIONS : 31 / 31 completed

Max Array

MAX ARRAY
Write a Java program to find the largest element in an array of integers using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         int a=sc.nextInt();
6         int[] arr=new int[a];
7         for(int i=0;i<a;i++){
8             arr[i]=sc.nextInt();
9         }
10        int max=arr[0];
11        for(int i=1;i<a;i++){
12            if(max<arr[i]){
13                max=arr[i];
14            }
15        }
16        System.out.println("maximum element is: "+max);
17    }
18 }
```

OUTPUT

maximum element is: 4

INPUT

4
1
2
3
4

27. 2nd Max

QUESTIONS 31 / 31 completed

2nd Max

2ND MAX

Write a Java program to find the second largest element in an array of integers using loops and conditional statements.

PROGRAM EDITOR

Java Run Save

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         int a=sc.nextInt();
6         int[] arr=new int[a];
7         for(int i=0;i<a;i++){
8             arr[i]=sc.nextInt();
9         }
10        for(int i=0;i<a-1;i++){
11            for(int j=i+1;j<a;j++){
12                if(arr[i]>arr[j]){
13                    int temp=arr[i];
14                    arr[i]=arr[j];
15                    arr[j]=temp;
16                }
17            }
18        }
19        System.out.println(arr[a-2]);
20    }
21 }
```

OUTPUT

4

INPUT

5
1
2
3
4
5

28. Min

QUESTIONS 31 / 31 completed

Min

MIN

Write a Java program to find the smallest element in an array of integers using loops and conditional statements.

PROGRAM EDITOR

Java Run Save

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         int a=sc.nextInt();
6         int[] arr=new int[a];
7         for(int i=0;i<a;i++){
8             arr[i]=sc.nextInt();
9         }
10        int min=arr[0];
11        for(int i=0;i<a;i++){
12            if(min>arr[i]){
13                min=arr[i];
14            }
15        }
16        System.out.println("Minimum element: "+min);
17    }
18 }
```

OUTPUT

Minimum element: 2

INPUT

5
6
2
3
4
5

29. 2nd Min

QUESTIONS 31 / 31 completed

2nd Min

2ND MIN

Write a Java program to find the second smallest element in an array of integers using loops and conditional statements.

PROGRAM EDITOR

Java Run Save

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         int a=sc.nextInt();
6         int[] arr=new int[a];
7         for(int i=0;i<a;i++){
8             arr[i]=sc.nextInt();
9         }
10        for(int i=0;i<a-1;i++){
11            for(int j=i+1;j<a;j++){
12                if(arr[i]>arr[j]){
13                    int temp=arr[i];
14                    arr[i]=arr[j];
15                    arr[j]=temp;
16                }
17            }
18        }
19        System.out.println("The Second Smallest Element is: "+arr[1]);
20    }
21 }
```

OUTPUT

The Second Smallest Element is: 5

INPUT

5
12
6
11
4
5

30. Bubble Sort

QUESTIONS 31 / 31 completed

Sort-B

SORT-B

Write a Java program to implement bubble sort on an array of integers using loops and conditional statements.

PROGRAM EDITOR

Java Run Save

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         int a=sc.nextInt();
6         int[] arr=new int[a];
7         for(int i=0;i<a;i++){
8             arr[i]=sc.nextInt();
9         }
10        for(int i=0;i<a-1;i++){
11            for(int j=0;j<a-i-1;j++){
12                if(arr[j]>arr[j+1]){
13                    int temp=arr[j];
14                    arr[j]=arr[j+1];
15                    arr[j+1]=temp;
16                }
17            }
18        }
19        System.out.println("Sorted Order: ");
20        for(int i=0;i<a;i++){
21            System.out.print(arr[i]+" ");
22        }
23    }
24 }
```

OUTPUT

Sorted Order:
1 2 3 4 5

INPUT

5
5
4
3
2
1

31. Insertion Sort

QUESTIONS : 31 / 31 completed

Sort-I

SORT-I

Write a Java program to implement insertion sort on an array of integers using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         int a=sc.nextInt();
6         int[] arr=new int[a];
7         for(int i=0;i<a;i++){
8             arr[i]=sc.nextInt();
9         }
10        for(int i=1;i<a;i++){
11            int key=arr[i];
12            int j=i-1;
13            while(j>0 && arr[j]>key){
14                arr[j+1]=arr[j];
15                j--;
16            }
17            arr[j+1]=key;
18        }
19        System.out.println("Sorted Order: ");
20        for(int i=0;i<a;i++){
21            System.out.print(arr[i]+" ");
22        }
23    }
24 }
```

OUTPUT

```
Sorted Order:
3 4 5 6 7
```

INPUT

```
5
4
5
3
7
6
```